

15 A At displacement = $\frac{A}{2}$, $u^2 = \omega^2 \left[A^2 - \left(\frac{A^2}{4} \right) \right]$

At displacement = 0, $v^2 = \omega^2 A^2$

Therefore $v = \frac{2}{\sqrt{3}} u$

16 B The oscillation experiences greater damping in oil than in water. Maximum amplitude